Objectives

- Explain React components
- Identify the differences between components and JavaScript functions
- Identify the types of components
- Explain class component
- Explain function component
- Define component constructor
- Define render() function

In this hands-on lab, you will learn how to:

- Create a function component
- Apply style to components
- Render a component

Prerequisites

The following is required to complete this hands-on lab:

- Node.js
- NPM
- Visual Studio Code

Notes

Estimated time to complete this lab: **30 minutes.**

Create a react app for Student Management Portal named scorecalculatorapp and create a function component named “CalculateScore” which will accept Name, School, Total and goal in order to calculate the average score of a student and display the same.

1. Create a React project named “scorecalculatorapp” type the following command in terminal of Visual studio:

   ```
   C:>npx create-react-app scorecalculatorapp
   ```

2. Create a new folder under Src folder with the name “Components”. Add a new file named “CalculateScore.js”

3. Type the following code in CalculateScore.js

```
import '../Stylesheets/mystyle.css'

const percentToDecimal= (decimal) => {
    return (decimal.toFixed(2) + '%')
}

const calcScore = (total, goal) => {
    return percentToDecimal(total/goal)
}

export const CalculateScore = ({Name,School, total, goal}) => (
        <div className="formatstyle">
            <h1><font color="Brown">Student Details:</font></h1>
            <div className="Name">
               <b> <span> Name:  </span> </b>
                <span>{Name}</span>
            </div>
            <div className="School">
               <b> <span> School: </span> </b>
                <span>{School}</span>
            </div>
            <div className="Total">
                <b><span>Total:</span> </b>
                <span>{total}</span>
                <span>Marks</span>
            </div>
            <div className="Score">
                <b>Score:</b>
                <span>
                    {calcScore(
                        total,
                        goal
                    )}
                </span>
            </div>
        </div>
```

4. Create a Folder named Stylesheets and add a file named “mystyle.css” in order to add some styles to the components:

```css
.Name
{
  font-weight:300;
 color:blue;
}
.School
{
 color:crimson;
}
.Total
{
 color:darkmagenta;
}
.formatstyle
{
   text-align:center;
   font-size:large;
}
.Score
{
   color:forestgreen;
}
```

5. Edit the App.js to invoke the CalculateScore functional component as follows:

```javascript
import {CalculateScore} from '../src/components/CalculateScore';

function App()
{

return(
<div>
   <CalculateScore Name={"Steeve"}
   School={"DNV Public School"}
   total={284}
   goal={3}
   />
</div>
)
}

export default App;
```

6. In command Prompt, navigate into scorecalculatorapp and execute the code by typing the following command:

```
C:\scorecalculatorapp>npm start
```

7. Open browser and type “localhost:3000” in the address bar:

Windows PowerShell

```
  cd scorecalculatorapp
  npm start

Happy hacking!
PS C:\Users\anana\Downloads\Digital Nurture\Week 5\React\3. ReactJS-HOL> cd scorecalculatorapp
PS C:\Users\anana\Downloads\Digital Nurture\Week 5\React\3. ReactJS-HOL\scorecalculatorapp> npm start

> scorecalculatorapp@0.1.0 start
> react-scripts start

(node:29240) [DEP0176] DeprecationWarning: fs.F_OK is deprecated, use fs.constants.F_OK instead
(Use `node --trace-deprecation ...` to show where the warning was created)
(node:29240) [DEP_WEBPACK_DEV_SERVER_ON_AFTER_SETUP_MIDDLEWARE] DeprecationWarning: 'onAfterSetupMiddleware' option is d
eprecated. Please use the 'setupMiddlewares' option.
(node:29240) [DEP_WEBPACK_DEV_SERVER_ON_BEFORE_SETUP_MIDDLEWARE] DeprecationWarning: 'onBeforeSetupMiddleware' option is
 deprecated. Please use the 'setupMiddlewares' option.
Starting the development server...
Compiled successfully!

You can now view scorecalculatorapp in the browser.

  Local:            http://localhost:3000
  On Your Network:  http://192.168.0.87:3000

Note that the development build is not optimized.
To create a production build, use npm run build.

webpack compiled successfully
```

mystyle.css | CalculateScore.js | App.js

src > App.js > default

```
import { CalculateScore } from './components/CalculateScore';

function App() {
  return (
    <div>
      <CalculateScore
        Name={"Steeve"}
        School={"DNV Public School"}
        total={284}
        goal={3}
      />
    </div>
  );
}

export default App;
```

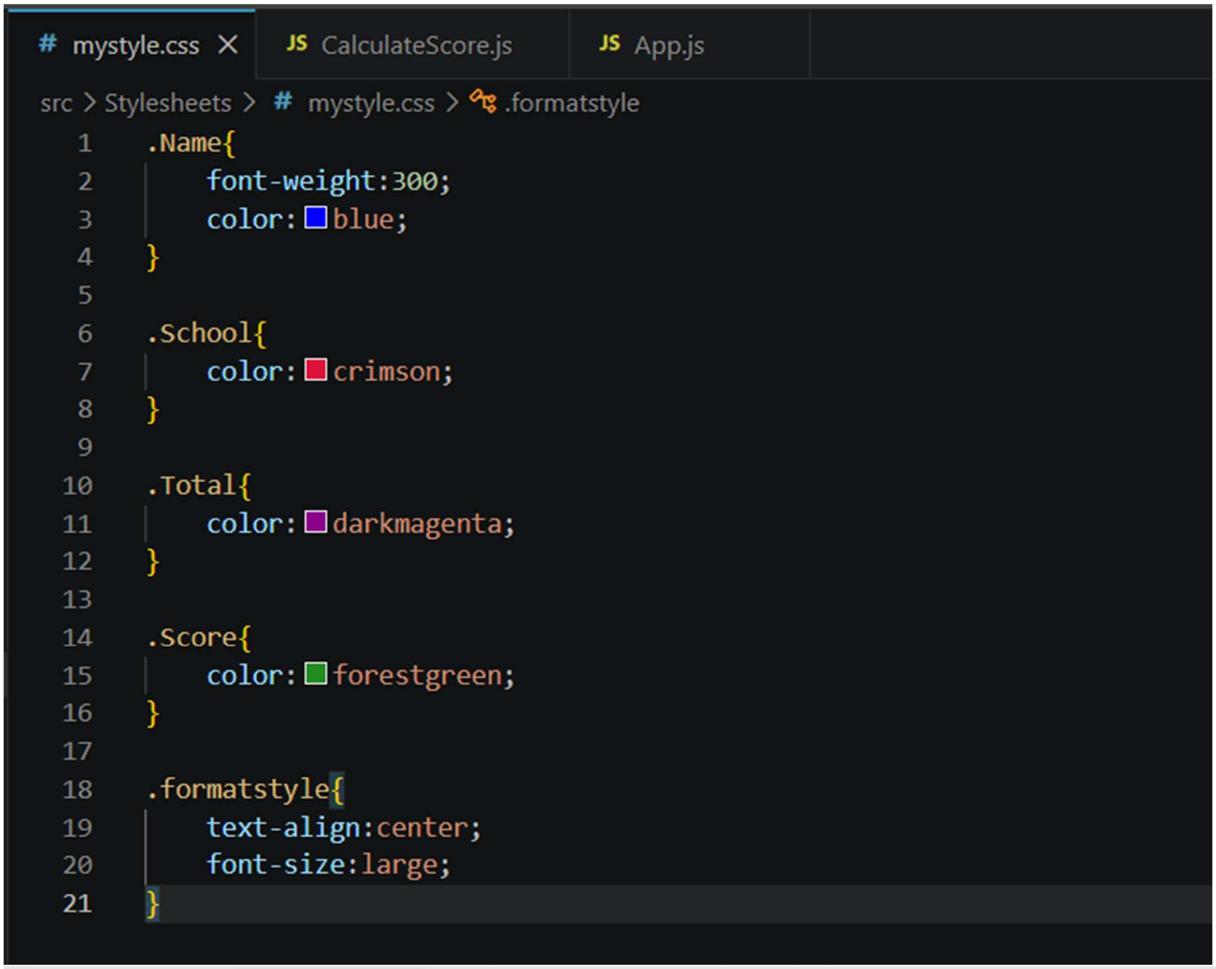
CHAT

SESSI

mystyle.css | CalculateScore.js | App.js

src > components > CalculateScore.js > CalculateScore

```
const percentToDecimal = (decimal) => {
}

const calcScore = (total, goal) => {
    return percentToDecimal(total / goal);
}

export const CalculateScore = ({ Name, School, total, goal }) => (
    <div className="formatstyle">

        <h1>
            <font color="Brown">Student Details:</font>
        </h1>

        <div className="Name">
            <b><span>Name: </span></b>
            <span>{Name}</span>
        </div>

        <div className="School">
            <b><span>School: </span></b>
            <span>{School}</span>
        </div>

        <div className="Total">
            <b><span>Total: </span></b>
            <span>{total}</span>
            <span> Marks</span>
        </div>

        <div className="Score">
            <b>Score: </b>
            <span>{calcScore(total, goal)}</span>
        </div>

    </div>
```




```css
.Name{
    font-weight:300;
    color:blue;
}

.School{
    color:crimson;
}

.Total{
    color:darkmagenta;
}

.Score{
    color:forestgreen;
}

.formatstyle{
    text-align:center;
    font-size:large;
}
```